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Atmospheric Research 37 (1995) 247–256

ATMOSPHERIC
RESEARCH

Atmospheric ozone as a climate gas

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Accepted 17 November 1994

Abstract

Observations during the past few decades indicate that the distribution of atmospheric ozone has been changing, notably decreases in the lower stratosphere around the globe and increases in the upper troposphere in the mid-to-high latitudes of the Northern Hemisphere. These changes can perturb the radiative forcing of the Earth–atmosphere climate system and thus raise the issue of atmospheric ozone as a climate gas. Here, we demonstrate the importance of ozone to climate by comparing general circulation model (GCM) simulated present climates between two atmospheric ozone climatologies: a version currently used in the NCAR GCM's (CCM1 and GENESIS) and an updated version. The new ozone climatology includes two improvements: the use of recent years' measurements from satellite (TOMS and SAGE) and ground-based ozonesondes and the consideration of longitudinal variations. Although the effect on the simulated global, annual mean surface air temperature is calculated to be small, the regional influences are quite substantial, especially at the middle and high latitudes of the Northern Hemisphere during winter. This suggests that atmospheric ozone is important in simulating the regional climate and observed ozone changes in the last few decades may have played an important role in affecting the climate. Note that to assess the climatic effect of ozone changes requires the GCM's to include interactive radiation–chemistry–dynamics interaction so that the ozone distribution can be consistently calculated.

1. Introduction

Atmospheric ozone (O_3) absorbs solar radiation, in particular, the UV-B radiation. It also traps outgoing longwave radiation similar to greenhouse gases H_2O , CO_2 , CH_4 , CFC's and N_2O . The presence of O_3 in the Earth's atmosphere is therefore important not only to the biological activities on the surface but also the radiation budget and climate (Bower and Ward, 1982). Consequently, realistic representation of O_3 spatial distribution in GCM's

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is needed to simulate the present climate. In addition, one of the major findings from the Intergovernmental Panel on Climate Change (IPCC), (Houghton et al., 1992) and the Scientific Assessment of Ozone Depletion (WMO, 1991) was the identification of possible climatic effects over the last few decades resulting from anthropogenically-induced changes in atmospheric O_3 .

The effects of changes in atmospheric O_3 on the solar and longwave radiative forcing of the troposphere–surface system are sensitive to the altitudes where O_3 changes. For example, a decrease in stratospheric O_3 will increase the radiative forcing (a warming effect) due to increased available solar radiation for absorption, and decrease the radiative forcing (a cooling effect) due to decreased downward longwave radiation. The net compensating effect will depend on the location and time of the year. On the other hand, an increase in tropospheric O_3 can warm the troposphere–surface system through increases in absorption of both the solar and longwave radiation. Sensitivity calculations indicate that the surface temperature is most sensitive to O_3 changes in the lower stratosphere and upper troposphere (Wang et al., 1980; Lacis et al., 1990). Consequently, study of the climatic effects of O_3 changes focuses on the observed O_3 changes in these regions.

Calculations of the climatic impact of lower stratospheric O_3 decreases, which have been observed over the last decade, was presented in Houghton et al. (1992). It was clearly demonstrated that the observed O_3 changes could lead to significant impact on the radiative forcing, particularly at high latitudes. The effect has been to reduce the radiative forcing, thereby reducing the expected impact of the other greenhouse gases. However, the results are sensitive to the temperature responses assumed in the lower stratosphere in responding to O_3 decreases. On the other hand, the few available ozonesondes indicate that the upper tropospheric O_3 has been increasing at the mid- and high-latitude of the Northern Hemisphere during the last few decades (Mohnen et al., 1993; Wang et al., 1993). Calculations of the changes in radiative forcing from these observed increases during the 1980's (versus the 1970's) give changes which are comparable to the changes from the other greenhouse gases during the same period.

The calculated impact on the radiative forcing from both lower stratospheric depletion and tropospheric increases clearly demonstrates the potential importance of atmospheric O_3 as a climate gas. Because of the highly inhomogeneous distribution of O_3 , GCM's are needed to study the climatic effects. However, before conducting the GCM climate perturbation experiments, the zonally symmetric O_3 climatology (Dutsch, 1978; below referred as the old O_3 data) of the NCAR GCM's (CCM1 and GENESIS), which we used for climate studies (Wang et al., 1991a, 1992), needs to be updated with more recent observations and to include the longitudinal variation. In this note, we present results of climate simulation using a new O_3 climatology based on available satellite and ground-based ozonesonde measurements; the climatology has monthly-mean values as a function of latitude, longitude, and altitude constructed. Section 2 describes the new O_3 data. Comparison of the climate simulations between the new and old O_3 climatologies is discussed in Section 3 and conclusions are given in Section 4.

2. Atmospheric O_3 climatology

In GCM's, the O_3 climatology needs to specify both the total column amount and the vertical distribution. The total column O_3 is taken from Total Ozone Mapping Spectrometer

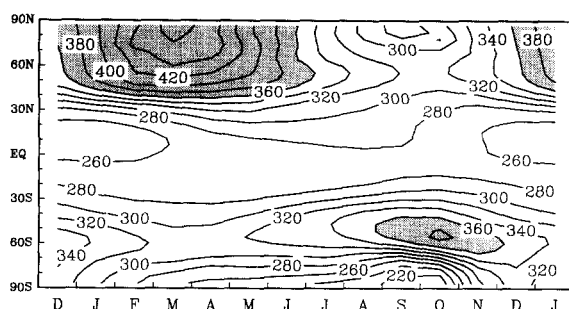


Fig. 1. Latitude-monthly mean total column ozone (in Dobson Unit, 10^{-3} cm). The ozone climatology is based on TOMS averaged over 11/1978–1/1992 with missing data at high latitudes interpolated and verified against available ground-based measurements (WMO, 1993).

(TOMS) measurements which have monthly-mean values from November 1978 to January 1992 with resolution of 1.25° longitude and 2° latitude (Bowman and Krueger, 1985; Stolarski et al., 1991). The missing values at high latitudes in both Hemispheres, mostly during winter, were filled with available observations near the poles using an interpolation scheme that yields consistent seasonal and interannual variations. These values are in excellent agreement with measured values available over the Arctic region (WMO, 1993).

Fig. 1 shows the monthly mean values of column O_3 amount averaged over the TOMS measuring period. The important features include: low values at low latitudes throughout the year, spring maximum at middle and high latitudes with much larger magnitudes in the Northern Hemisphere, and the Antarctic O_3 hole during September–October. Comparison of total O_3 column between the new and old climatologies indicate that the difference is generally small except during late summer and early fall in the areas of south of $70^\circ S$ where the new data is much smaller because of the Antarctic Ozone hole and, from May through November in low and middle latitudes where the new data is larger by 10–15%.

Note that, in addition to the difference in the zonal mean values between the new and old O_3 climatologies, the old data does not include the longitudinal contrast, which can be substantial as shown in Fig. 2 for January. The longitudinal variation is small in low latitudes and most of the Southern Hemisphere. However, the contrast is substantial North of $30^\circ N$; for example, at $50^\circ N$ the value has a minimum ~ 340 DU in Northern Europe versus a maximum 460 DU in Northern Japan. This contrast can be clearly seen in the comparison between the new and old climatologies. For example, the new data is smaller over Northern Europe, but larger over Japan and Northern China.

As mentioned earlier, the vertical distribution of O_3 is important in determining the radiative forcing. It needs to be realistically prescribed, and ultimately calculated with interactive chemistry and climate. In the new O_3 climatology, the stratospheric distribution is based on the data from the Stratospheric Aerosol and Gas Experiment (SAGE-II) which provides data up to 60 km at 1-km resolution for the period October 1984 to November 1989 (Cunnold et al., 1989; McCormick et al., 1992). Above 60 km, a single mean value at 100 km taken from McClatchey et al. (1971) is used to calculate the mixing ratios assuming linear relationship between $\log(O_3)$ and pressure. For the troposphere, we used the data from Logan (1985) and Spivakovsky et al. (1990), which has values at 1000, 900,

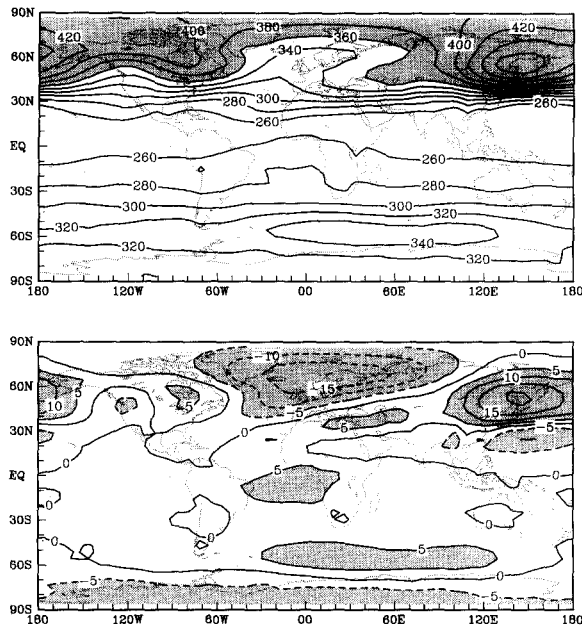


Fig. 2. January, latitude–longitudinal distribution of (a) new column ozone (Dobson Unit, 10^{-3} cm) and (b) difference (%) between new and old ozone climatologies

850, 800, 700, 500, 300, 200, 150 and 70 mb with 10 to 20 degree latitude resolution. Again linear interpolation is used to calculate the values between 12 and 17 km. As an example, Fig. 3 shows the January, altitude–latitudinal distribution of the O_3 mixing ratio. There is a hemispheric symmetry with maximum value of about 10 ppmv around 32 km at the equatorial region. The values in the troposphere are generally much smaller than those in the stratosphere and total tropospheric column is about 10% of the total column shown in Fig. 1. Differences between the new and old O_3 January climatologies are also shown in Fig. 3. In general, the new data is larger than the old mainly in the Northern Hemisphere; the difference is particularly large in the regions of mid-to-upper troposphere and lower stratosphere at low latitudes. A combination of the difference in the total column and vertical distribution between the new and old O_3 data can yield different model climates.

3. Climate simulations

The climate simulations reported here is based on the use of the GENESIS (General Environmental and Ecological Simulation of Interactive Systems) global climate model (Thompson and Pollard, 1995) for the atmosphere and a 50-meter mixed layer with prescribed heat transport for the ocean. Note that the longwave radiation parameterization of Wang et al. (1991b) is used. The AGCM originated from the NCAR Community Climate Model version 1 (CCM1), but nearly all components have been replaced or extensively modified with the exception of the spectral dynamics. This version uses R15/L12 resolution

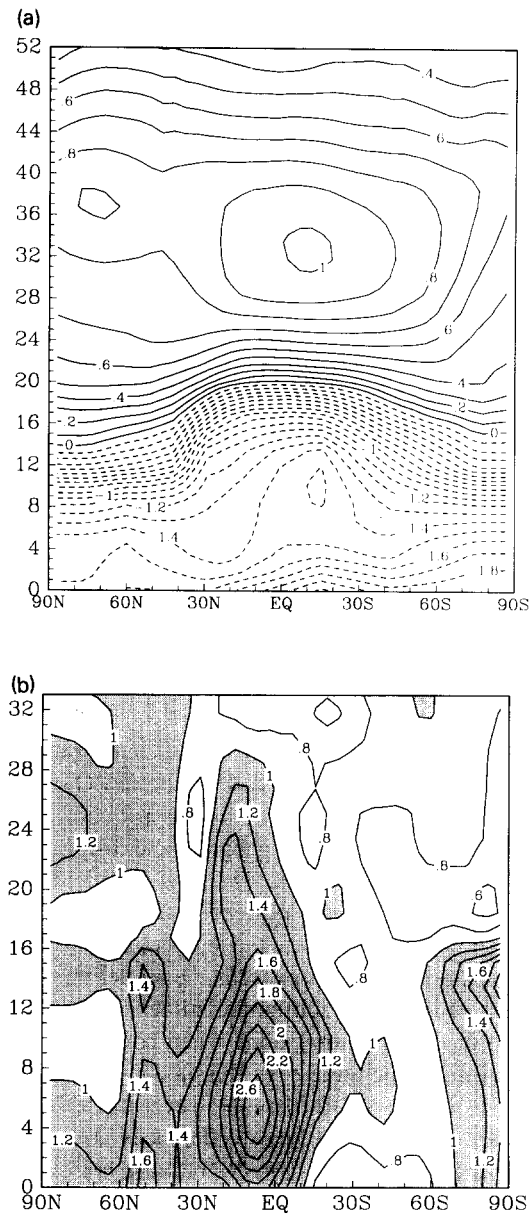


Fig. 3. January, altitude–latitudinal ozone distribution of (a) new ozone climatology (ppmv with logarithmic scale) and (b) ratio of new to old data.

with a diurnal cycle and has improved physics in radiation, clouds, boundary layer and convection; the ground includes a 3-layer snow and a 6-layer soil model. We run two model simulations: one with the old O_3 climatology and the other with the new O_3 climatology. For each simulation, a period of twenty-years is run and the last ten-years equilibrium climate statistics are used for analysis and comparison.

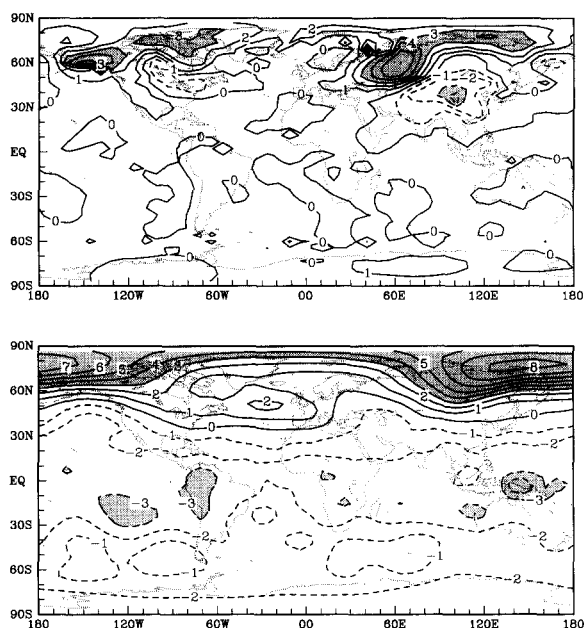


Fig. 4. Model simulated temperature difference ($^{\circ}\text{C}$) at (a) surface and (b) 60 mb level between new and old ozone climatologies.

The difference in the global, annual mean surface air temperature between the two O_3 climatologies is calculated to be very small, 288.41 K for new O_3 and 288.57 K for old O_3 . However, differences in the seasonal and longitudinal distribution is quite large. For example, Fig. 4 shows the differences in January, geographical distribution of surface air temperature between the two O_3 climatologies. The new O_3 yields a warmer surface air temperature for most regions and the warming effect is particularly large in the areas centered around (60E; 60N) and (130W; 60N) where the temperature can be higher by as much as 4–5 $^{\circ}\text{C}$. The magnitude of these warmings exceeds the model interannual variability, thus implying that atmospheric O_3 can have substantial influences on the regional climate and, for some areas, improve the model simulations when compared with observations. For example, the warmer surface temperature over Russia corrects substantially the cold bias (about 10 $^{\circ}\text{C}$) simulated using the old O_3 data. However, the effects are mixed for other middle and high latitude regions where large differences are calculated. Note that the warming effect is uniformly small at low latitudes because both the difference in column O_3 and the longitudinal contrast are small. There are two areas, North America and Central Eurasia, where cooler surface temperatures are calculated. As mentioned earlier, the surface warming or cooling depends on the net effect of changes in solar and longwave radiation, which is a function of latitude and season. Our analyses indicate that the warmer surface temperature is initially caused by the enhanced longwave greenhouse effect and the cooling by the decreased incident solar radiation and subsequently modified by the climate feedbacks.

The difference in the 60 mb temperature is also shown in Fig. 4. Warmer temperatures are calculated at the middle and high latitudes in the Northern Hemisphere while cooler

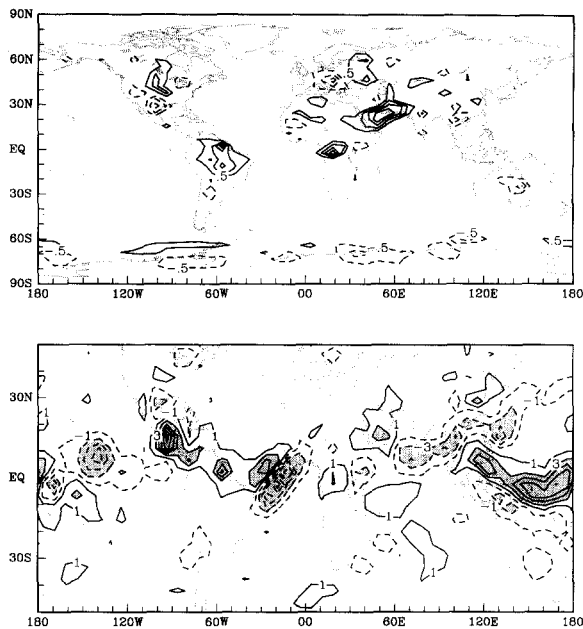


Fig. 5. Model simulated difference in July (a) diurnal range ($^{\circ}\text{C}$) of the surface air temperature and (b) precipitation (mm/day) between the new and old O_3 data.

temperatures are found for other regions. The temperature pattern is similar to the pattern of O_3 column shown in Fig. 1 because of the close association between O_3 and temperature in the lower stratosphere.

Because of the focus of this MINIMAX Workshop, we have also examined the effect of O_3 climatologies on the simulations of the diurnal temperature-range of the surface air temperature and the precipitation (and thus the cloudiness). Because of the large effect of O_3 on solar radiation (and thus the maximum temperature), the July results are examined and shown in Fig. 5. Between the new and old O_3 climatologies, large differences exist in temperature ranges in several low and middle latitude continental regions while the effect on the precipitation is mainly within the ITCZ. However, the uncertainties associated with these changes, in particular the precipitation, are large because that these changes are based on ten-years equilibrium simulations which are not long enough from statistical consideration and that GENESIS simulated precipitation patterns showed large interannual fluctuations within ITCZ. Nevertheless, it is clear that to use GCM's to study the observed characteristics of a reduced diurnal range associated with a larger increase of the minimum (than maximum) temperature will require the GCM's to include realistic distribution of atmospheric O_3 and its change as well as changes in other radiatively-important constituents such as greenhouse gases and sulfates.

The differences in the vertical distributions of zonal mean temperature and wind for January between the two O_3 climatologies are shown in Fig. 6. For temperature, very small difference is calculated in the troposphere. However, in the lower stratosphere, the new data calculates a warmer temperature in polar regions and a cooler temperature in the equatorial

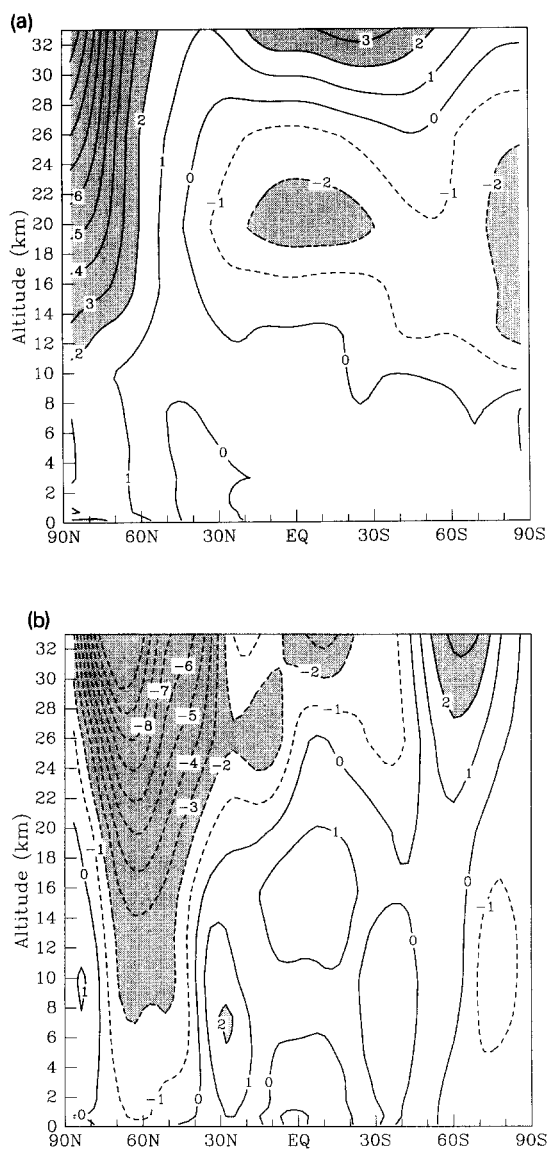


Fig. 6. Comparison of the January, altitude–latitudinal distribution of (a) temperature (°C) and (b) zonal wind (m/s) between new and old O₃ data.

region. Note that the warmer polar night lower stratosphere is a needed improvement since nearly all GCM's calculate cooler temperature when compare with observations (Boer et al., 1992). Difference in the zonal wind occurs mainly at the middle to high latitudes of the Northern Hemisphere. Again, the decrease is also an improvement.

4. Conclusions

The GCM simulations suggest that atmospheric ozone is important to regional climate. Because of its large seasonal variation, the effect of ozone is particularly large at the middle

to high latitudes. The differences in the longitudinal contrast between the two ozone climatologies yield large differences in the longitudinal surface temperature, which imply the observed ozone changes in the last few decades may have contributed regional climate changes. However, GCM studies with imposed ozone changes are required to quantify the climatic effect versus increases of other greenhouse gases CO₂, CH₄, N₂O, and CFC's (Houghton et al., 1992).

Finally, two considerations make ozone distinctly different from those other greenhouse gases which are uniformly-mixed in the troposphere. First, ozone is a secondary constituent formed by chemical reactions in the atmosphere. Second, anthropogenically-induced changes in the ozone distribution show large spatial and temporal variations in the atmosphere. Therefore, study of the climatic impact of ozone requires better resolution in its spatial representation and ultimately consideration of climate–chemistry interactions.

Acknowledgements

We thank J. Logan for useful discussion and for providing the tropospheric ozone distribution. The research was supported by the Environmental Science Division/Department of Energy and the Climate Dynamics/National Science Foundation. We appreciate the additional supercomputer time to carry out the GCM simulations at NCAR provided by the Model Evaluation Consortium for Climate Assessment (MECCA)/Electric Power Research Institute.

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